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## NYCU IMF - 114 Financial Econometrics

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**Assignment:** 5, Ch8 q6, 16

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### Question 6

Consider the wage equation

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i \quad (\text{XR8.6a})$$

where wage is measured in dollars per hour, education and experience are in years, and  $METRO = 1$  if the person lives in a metropolitan area. We have  $N = 1000$  observations from 2013.

- (a) We are curious whether holding education, experience, and METRO constant, there is the same amount of random variation in wages for males and females. Suppose  $\text{var}(e_i|x_i, FEMALE = 0) = \sigma_M^2$  and  $\text{var}(e_i|x_i, FEMALE = 1) = \sigma_F^2$ . We specifically wish to test the null hypothesis  $\sigma_M^2 = \sigma_F^2$  against  $\sigma_M^2 \neq \sigma_F^2$ . Using 577 observations on males, we obtain the sum of squared OLS residuals,  $SSE_M = 97161.9174$ . The regression using data on females yields  $\hat{\sigma}_F = 12.024$ . Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

To test if the error variance is the same for males and females, we use an  $F$ -test.

Hypotheses:  $H_0 : \sigma_M^2 = \sigma_F^2$  vs  $H_1 : \sigma_M^2 \neq \sigma_F^2$

Step 1: Find the variance for Males ( $\hat{\sigma}_M^2$ )

For males,  $N = 577$  and the model has 4 parameters ( $K = 4$ : intercept, EDUC, EXPER, METRO).

$$\hat{\sigma}_M^2 = \frac{SSE_M}{N - K} = \frac{97161.9174}{577 - 4} = \frac{97161.9174}{573} \approx 169.567046$$

Step 2: Find the variance for Females ( $\hat{\sigma}_F^2$ )

Given that  $\hat{\sigma}_F = 12.024$ . Square this to get the variance.

$$\hat{\sigma}_F^2 = (12.024)^2 \approx 144.576576$$

Step 3: Calculate the  $F$ -statistic

$$F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_F^2} \approx \frac{169.567046}{144.576576} \approx 1.1729$$

Step 4: Determine the Rejection Region

The degrees of freedom are  $df_1 = 573$  (numerator) and  $df_2 = (1000 - 577) - 4 = 419$

(denominator). Since it's a two-tailed test at the 5% level, the critical values are  $F_{0.025,573,419}$  and  $F_{0.975,573,419}$ .

Calculated Test Statistic:  $F \approx 1.173$

Lower Critical Value:  $F_{0.025,573,419} \approx 0.8377$

Upper Critical Value:  $F_{0.975,573,419} \approx 1.1968$

The Conclusion

Because the calculated  $F$ -statistic (1.1729) falls right inside the non-rejection region—between the lower bound (0.8377) and the upper bound (1.1968). Therefore, at the 5% significance level, we do not reject the null hypothesis ( $H_0$ ), there is not enough statistical evidence to conclude that the error variances for males and females are different.

- (b) We hypothesize that married individuals, relying on spousal support, can seek wider employment types and hence holding all else equal should have more variable wages. Suppose  $\text{var}(e_i|x_i, \text{MARRIED} = 0) = \sigma_{\text{SINGLE}}^2$  and  $\text{var}(e_i|x_i, \text{MARRIED} = 1) = \sigma_{\text{MARRIED}}^2$ . Specify the null hypothesis  $\sigma_{\text{SINGLE}}^2 = \sigma_{\text{MARRIED}}^2$  versus the alternative hypothesis  $\sigma_{\text{MARRIED}}^2 > \sigma_{\text{SINGLE}}^2$ . We add *FEMALE* to the wage equation as an explanatory variable, so that

$$\text{WAGE}_i = \beta_1 + \beta_2 \text{EDUC}_i + \beta_3 \text{EXPER}_i + \beta_4 \text{METRO}_i + \beta_5 \text{FEMALE}_i + e_i \quad (\text{XR8.6b})$$

Using  $N = 400$  observations on single individuals, OLS estimation of (XR8.6b) yields a sum of squared residuals is 56231.0382. For the 600 married individuals, the sum of squared errors is 100,703.0471. Test the null hypothesis at the 5% level of significance. Clearly state the value of the test statistic and the rejection region, along with your conclusion.

Hypotheses:  $H_0 : \sigma_{\text{SINGLE}}^2 = \sigma_{\text{MARRIED}}^2$  vs  $H_1 : \sigma_{\text{MARRIED}}^2 > \sigma_{\text{SINGLE}}^2$

Step 1: Calculate variances

$$\begin{aligned}\hat{\sigma}_{\text{SINGLE}}^2 &= \frac{56231.0382}{400 - 5} \approx 142.357059 \\ \hat{\sigma}_{\text{MARRIED}}^2 &= \frac{100,703.0471}{600 - 5} \approx 169.248819\end{aligned}$$

Step 2: Calculate the  $F$ -statistic

$$F = \frac{169.248819}{142.357059} \approx 1.188904$$

Step 3: Determine the Rejection Region

Since it's a one-tailed test at the 5% level, the critical values are  $F_{0.95,595,395}$ .

$$F_{0.95,595,395} = 1.164705$$

### The Conclusion

Because the calculated  $F$ -statistic (1.1889) exceeds the critical value (1.1647), it falls in the rejection region. Therefore, at the 5% significance level, we reject the null hypothesis ( $H_0$ ), there is sufficient statistical evidence to conclude that the error variance for married individuals is significantly greater than the error variance for single individuals. This indicates the presence of heteroskedasticity between these two groups.

- (c) Following the regression in part (b), we carry out the  $NR^2$  test using the right-hand-side variables in (XR8.6b) as candidates related to the heteroskedasticity. The value of this statistic is 59.03. What do we conclude about heteroskedasticity, at the 5% level? Does this provide evidence about the issue discussed in part (b), whether the error variation is different for married and unmarried individuals? Explain.

### 1. The Hypotheses

- $H_0$ : The variance of the error term is constant (Homoskedasticity). Mathematically, the coefficients on all the explanatory variables in the auxiliary regression are zero.
- $H_1$ : The variance of the error term is not constant, and depends on one or more of the explanatory variables (Heteroskedasticity).

### 2. The Auxiliary Regression

The auxiliary equation:

$$\hat{e}^2 = \alpha_1 + \alpha_2 EDUC + \alpha_3 EXPER + \alpha_4 METRO + \alpha_5 FEMALE + v$$

### 3. Degrees of Freedom ( $df$ )

The test statistic follows a Chi-square ( $\chi^2$ ) distribution. The degrees of freedom for this test equal the number of explanatory variables in the auxiliary regression, excluding the intercept. As the number of variables is 4, therefore,  $df = 4$ .

### 4. Critical Value

Given that the calculated test statistic:  $N \times R^2 = 59.03$ .

Critical Value ( $\chi^2_{0.05,4}$ ) = 9.487729

### 5. Conclusion

Because  $59.03 > 9.488$ , the test statistic falls deep into the rejection region. We reject the null hypothesis ( $H_0$ ). There is strong statistical evidence of heteroskedasticity related to the variables  $EDUC$ ,  $EXPER$ ,  $METRO$ , and  $FEMALE$ .

### 6. Does this test married vs. unmarried variance?

The question asks if this specific  $NR^2$  test provides evidence that the error variance depends on whether a person is married or single.

The Answer is No. Because the variable  $MARRIED$  was not included as an explanatory variable in the auxiliary regression model. The  $NR^2$  test only detects heteroskedasticity driven by the variables explicitly included in the auxiliary regression on the right-hand

side. If we wanted to test whether MARRIED caused heteroskedasticity using this method, we would have to include it in the auxiliary equation.

- (d) Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?

### 1. The Hypotheses

- $H_0$ : The variance of the error term is constant (Homoskedasticity). All coefficients in the auxiliary regression are zero.
- $H_1$ : The variance of the error term is not constant (Heteroskedasticity). At least one coefficient in the auxiliary regression is non-zero.

### 2. Calculating the Degrees of Freedom ( $df$ )

The test statistic follows a Chi-square ( $\chi^2$ ) distribution. The degrees of freedom equal the number of explanatory variables in the auxiliary regression (excluding the intercept).

We count the variables for the White test:

- Original Variables (4):  $EDUC, EXPER, METRO, FEMALE$
- Squared Variables (2):  $EDUC^2, EXPER^2$
- Cross-Products (6): We pair every variable with every other variable.
  1.  $EDUC \times EXPER$
  2.  $EDUC \times METRO$
  3.  $EDUC \times FEMALE$
  4.  $EXPER \times METRO$
  5.  $EXPER \times FEMALE$
  6.  $METRO \times FEMALE$

Total Degrees of Freedom:  $4 + 2 + 6 = 12$

### 3. Critical Value

Given that the calculated test statistic:  $N \times R^2 = 78.82$ .

Critical Value ( $\chi^2_{0.05,12}$ ) = 21.02607

### 4. Conclusion

Because the calculated test statistic (78.82) is much larger than the critical value (21.02607), it falls into the rejection region. Therefore, we reject the null hypothesis ( $H_0$ ). At the 5% significance level, the White test provides strong evidence of heteroskedasticity in the model. The variance of the error term depends on the variables, their squares, or their interactions.

- (e) The OLS fitted model from part (b), with usual and robust standard errors, is

$$\begin{array}{cccccc} \widehat{WAGE} = & -17.77 & +2.50EDUC & +0.23EXPER & +3.23METRO & -4.2FEMALE \\ (se) & (2.36) & (0.14) & (0.031) & (1.05) & (0.81) \\ (robse) & (2.50) & (0.16) & (0.029) & (0.84) & (0.80) \end{array}$$

For which coefficients have interval estimates gotten narrower? For which coefficients have interval estimates gotten wider? Is there an inconsistency in the results?

Narrower (Robust < Usual): *EXPER* (0.029 < 0.031), *METRO* (0.84 < 1.05), *FEMALE* (0.80 < 0.81).

Wider (Robust > Usual): *EDUC* (2.50 > 2.36).

No, there is no inconsistency in the results. Heteroskedasticity-robust standard errors can be either larger or smaller than OLS standard errors. OLS standard errors are simply biased and inconsistent when heteroskedasticity is present; they don't uniformly overstate or understate the true variance.

- (f) If we add *MARRIED* to the model in part (b), we find that its *t*-value using a White heteroskedasticity robust standard error is about 1.0. Does this conflict with, or is it compatible with, the result in (b) concerning heteroskedasticity? Explain.

A *t*-value of 1.0 means *MARRIED* is not statistically significant in explaining the conditional mean of *WAGE*. This does not conflict with Part (b). A variable can be a strong driver of the variance of the error term (heteroskedasticity) without having a significant direct effect on the expected value of the dependent variable.

### Question 16

A sample of 200 Chicago households was taken to investigate how far American households tend to travel when they take a vacation. Consider the model

$$MILES = \beta_1 + \beta_2 INCOME + \beta_3 AGE + \beta_4 KIDS + e$$

*MILES* is miles driven per year, *INCOME* is measured in \$1000 units, *AGE* is the average age of the adult members of the household, and *KIDS* is the number of children.

- (a) Use the data file *vacation* to estimate the model by OLS. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant.

The 95% confidence interval for the  $\beta_4$  coefficient (*KIDS*):

$$[-135.3298, -28.32302]$$

- (b) Plot the OLS residuals versus *INCOME* and *AGE*. Do you observe any patterns suggesting that heteroskedasticity is present?

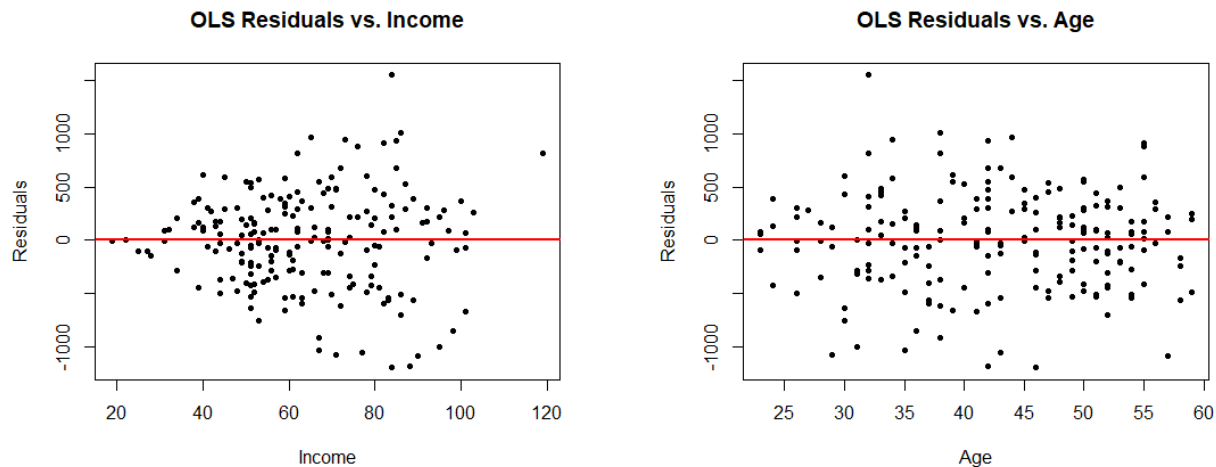


Figure 1: OLS residuals versus *INCOME* and *AGE*

In the Figure 1: OLS Residuals vs. Income, we can observe that as *INCOME* increases (moving from left to right on the x-axis), the vertical spread of the residuals becomes significantly wider. At low income levels (around 20-40), the errors are tightly clustered near the zero line. At higher income levels (80-120), the errors spread out wildly, ranging from roughly -1000 to +1000. This creates a classic "funnel" or "cone" shape. This pattern strongly suggests that heteroskedasticity is present, and specifically, that the variance of the error term is an increasing function of *INCOME*.

And in the Figure 1: OLS Residuals vs. Age, we can observe that as *AGE* increases, the vertical spread of the residuals remains relatively constant. The data points form a fairly uniform band across the graph from left to right. There is no distinct funneling or systematic pattern here. This suggests that the error variance does not depend on *AGE*.

To conclude, yes, we observe patterns suggesting that heteroskedasticity is present. Specifically, the plot of residuals against *INCOME* displays a clear widening variance, indicating that wealthier households have much more variation in their vacation travel miles compared to lower-income households.

- (c) Sort the data according to increasing magnitude of income. Estimate the model using the first 90 observations and again using the last 90 observations. Carry out the Goldfeld–Quandt test for heteroskedastic errors at the 5% level. State the null and alternative hypotheses.

We are testing if the error variance for the high-income group is greater than the error variance for the low-income group.

$$H_0: \sigma_{low}^2 = \sigma_{high}^2 \text{ vs } H_1: \sigma_{high}^2 > \sigma_{low}^2$$

First, we sort the data, run the regressions on the subsets, and calculate the  $F$ -statistic.

$$F = 3.104061$$

Then we find the Rejection Region, both subsets have  $N = 90$  and  $K = 4$  parameters. The degrees of freedom for both the numerator and denominator are  $N - K = 86$ .

$$F_{0.95,86,86} = 1.428617$$

The Conclusion

Because the calculated  $F$ -statistic (3.104061) exceeds the critical value (1.428617), it falls in the rejection region. Therefore, at the 5% significance level, we reject the null hypothesis ( $H_0$ ), there is sufficient statistical evidence to conclude that heteroskedasticity is present.

- (d) Estimate the model by OLS using heteroskedasticity robust standard errors. Construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How does this interval estimate compare to the one in (a)?

Comparing Part (d) to Part (a)

- Standard OLS Interval (Part a):  $[-135.33, -28.32]$  (Width  $\approx 107.01$ )
- Robust Interval (Part d):  $[-139.32, -24.33]$  (Width  $\approx 114.99$ )

The robust confidence interval is wider than the standard OLS interval. Because the variance of the errors expands at higher income levels (as seen in Part (b) plot), the standard OLS method underestimated the true standard errors. The robust calculation corrects this bias, widening the interval to accurately reflect the true uncertainty in the  $KIDS$  coefficient.

- (e) Obtain GLS estimates assuming  $\sigma_i^2 = \sigma^2 INCOME_i^2$ . Using both conventional GLS and robust GLS standard errors, construct a 95% interval estimate for the effect of one more child on miles traveled, holding the two other variables constant. How do these interval estimates compare to the ones in (a) and (d)?

Let's look at the width of the interval for the  $KIDS$  coefficient across three methods:

- Part (a) Standard OLS:  $[-135.33, -28.32]$ 
  - Width: 107.01
- Part (d) Robust OLS:  $[-139.32, -24.33]$ 
  - Width: 114.99
- Part (e) GLS (Weighted):  $[-119.89, -33.72]$ 
  - Width: 86.17

The Conclusion

The GLS interval is significantly narrower than both the standard OLS interval and the robust OLS interval.